

Milestone 2 Report: IAT 461

Regarding progress, I have identified the type of problem to solve. Based on the client's interest in finding aspects of fraudulent transactions to be able to determine if a new transaction is fraudulent or not, I have decided that this is a **structured** problem.

Data cleaning was attempted by checking for missing values in `merchant_zip_code`. I decided to drop those after noticing that there was a roughly equal proportion of such missing values within fraud and non-fraud transactions. I also checked the data for numerical columns with values that would not make sense to have as 0 or below, and have found that they don't exist. There is a column titled "Unnamed: 0" which appears to have been brought about by the way the dataset was exported - I tried to use `index_col=0` in `read_csv()` as a way to remove this column, but that hasn't worked - this column is inoffensive on its own, but it would be nice to figure out how to take it out.

Regarding EDA, because we want to determine the best way to predict if a transaction is fraud or not, it made sense to me to perform EDA on both the whole dataset and the dataset after being split into two portions: one with fraud and one without fraud, using the column `is_fraud` to make the split (it is a binary variable). If we can identify patterns in recurring values within certain classes, we could potentially use those as hardstop identifiers of whether or not a transaction is fraudulent. Thus, I identified categorical variables in `Transaction_Type`, `Payment_Method` and `Merchant_Category`, printing bar graphs to see if any particular value in each was found much more than other categories of the same column. However, I noticed that the distribution of categories in both fraud/non-fraud datasets was quite closely split - no one value was found in significantly higher amounts, but I did notice that there were winners edging out slightly over the rest in Groceries and Credit Card for `Transaction_Type` and `Merchant_Category`, respectively. Still, that is not enough evidence to say those values are a major tell for fraud.

Regarding numerical variables, taking the hint of Option B which suggests to determine frauds based on thresholds in `amt`, I created a boxplot for the `amt` variable - the amount of money involved in a transaction to see the differences in money involved in fraudulent and non-fraudulent transactions, and found that `amt` stayed squarely below 2000 (no currency was provided) across frauds, while non-frauds had many outliers stretching far beyond the amount of any fraud in the dataset. Therefore, I noted that a potential tell of a fraud is in transactions keeping within that under 2000 range.

I did want to follow up on that thought however by using a correlation matrix involving it, `is_fraud`, and other numerical variables: `'Customer_Age'`, `'Customer_Satisfaction_Score'` and `'Loyalty_Points_Earned'` to see if there are any notable correlations to `is_fraud`. `amt` was the only value with a correlation that stood out above the rest, at a positive correlation (0.26), while others were at 0.01 or even (-)0.00. While that correlation isn't high, it is still notable enough of a value to keep in mind moving forward as a potential tell.

While I have met the minimum number of visualizations (3) outlined by the client, I believe there is still more EDA to be done to identify relevant clues as to what makes a transaction fraudulent, as the lone correlation I have found could be supported by other correlated variables, and its correlation isn't that strong to begin with. I don't have enough information to make a decision regarding which option to go with out of the three the client presented, but by diving deeper into other variables such as location via zip code, cities, etc., and adding more correlation matrices for variables like Transaction_Type and Merchant_Category which have values slightly beating others in count, I might be able to find more leads.